



CEMENT – PROPERTIES, TYPES, USES & TESTS

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Cement - Definition

- ❖ Cement is a finely powdered binding material that sets and hardens when mixed with water.
- ❖ It is manufactured by burning a properly proportioned mixture of calcareous and argillaceous materials to form clinker, followed by grinding with gypsum.
- ❖ On adding water, cement undergoes hydration, forming binding products, which provides strength.
- ❖ Cement is mainly used as a binding material in concrete and mortar.
- ❖ Its typical grey colour is mainly due to iron compounds and other mineral constituents present in the raw materials.





Role of Cement in Concrete/Mortar

- ❖ Acts as the primary binding material that holds the constituents together.
- ❖ Reacts with water through hydration and forms C-S-H gel, which provides strength.
- ❖ In concrete, it binds fine and coarse aggregates into a dense mass.
- ❖ In mortar, it binds sand particles and provides cohesion.
- ❖ Fills voids between particles and contributes to strength, durability and impermeability.
- ❖ Provides the required workability and setting characteristics when properly proportioned.



Ideal Properties of Fresh Cement

- ❖ It should be easy to mix, handle, and apply.
- ❖ It should have a smooth and uniform consistency.
- ❖ It should remain workable for sufficient time before setting.
- ❖ It should possess good plasticity for easy spreading and application.
- ❖ It should retain sufficient water for proper hydration.
- ❖ It should have good adhesion with bricks, stones, and other materials.
- ❖ It should be free from lumps and impurities.
- ❖ It should have a uniform colour.
- ❖ It should not show segregation of its constituents.



OPC Grades

Grade	Meaning of Grade	Characteristic Compressive Strength	Typical Applications
OPC 33 Grade	Minimum 28-day compressive strength = 33 MPa	33 N/mm ²	Masonry, plastering, flooring and general low-strength works
OPC 43 Grade	Minimum 28-day compressive strength = 43 MPa	43 N/mm ²	General RCC, buildings, bridges, roads and precast works
OPC 53 Grade	Minimum 28-day compressive strength = 53 MPa	53 N/mm ²	High-strength RCC, prestressed concrete, high-rise structures and precast elements

Approximate market price (50 kg bag): OPC 33 – ₹350–400; OPC 43 – ₹360–420; OPC 53 – ₹375–450.



Chemical Constituents of OPC

Constituent	Typical range (%)	Role in OPC
Lime (CaO)	60–67	Forms calcium silicates; mainly responsible for strength . Excess causes unsoundness .
Silica (SiO₂)	17–25	Forms C₃S and C₂S ; contributes to strength and durability , especially at later ages.
Alumina (Al₂O₃)	3–8	Helps in clinker formation and promotes early setting; excess reduces strength and durability.
Iron oxide (Fe₂O₃)	0.5–6	Forms C₄AF , acts as a flux during burning and contributes to the grey colour of cement.
Magnesia (MgO)	0.1–4	Present in small amounts; contributes slightly to hardness, but excess causes expansion and unsoundness .
Sulphur trioxide (SO₃)	1–3	Mainly controls setting time through reaction with C ₃ A in the presence of gypsum; excess causes expansion.
Alkalis (Na₂O + K₂O)	0.2–1.3	Present in small quantities; excess may cause alkali–silica reaction (ASR) and efflorescence.



Bogues Compound

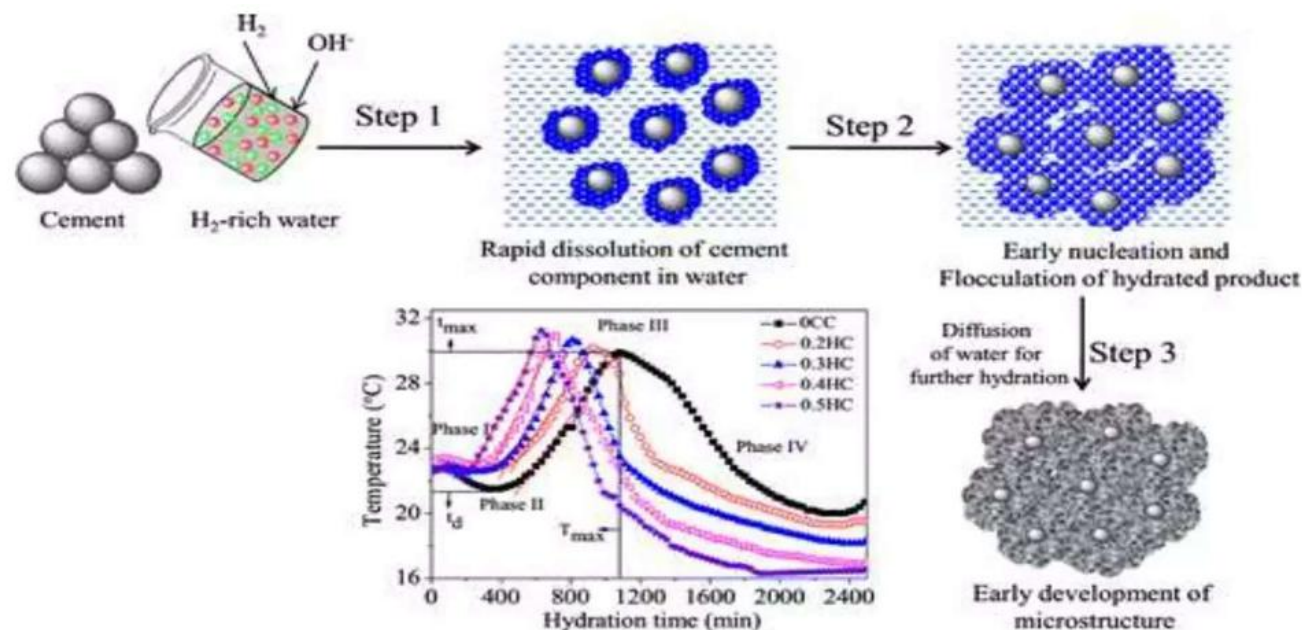
Compound	Formation	Contribution to Strength	Heat of Hydration	Effect on Setting
C_3S – Tricalcium silicate	$CaO + SiO_2 \rightarrow C_3S$	High early strength, mainly up to 28 days	High	Moderate contribution; hydrates rapidly
C_2S – Dicalcium silicate	$CaO + SiO_2 \rightarrow C_2S$	Later-age strength, develops slowly	Low	Slow hydration; little effect on early setting
C_3A – Tricalcium aluminate	$CaO + Al_2O_3 \rightarrow C_3A$	Very little direct strength contribution	Very high	Causes rapid setting if gypsum is absent; gypsum controls this reaction
C_4AF – Tetracalcium aluminoferrite	$CaO + Al_2O_3 + Fe_2O_3 \rightarrow C_4AF$	Small contribution to strength	Moderate	Relatively minor effect on setting

The symbols are cement chemist notation: C = CaO, S = SiO₂, A = Al₂O₃ and F = Fe₂O₃.



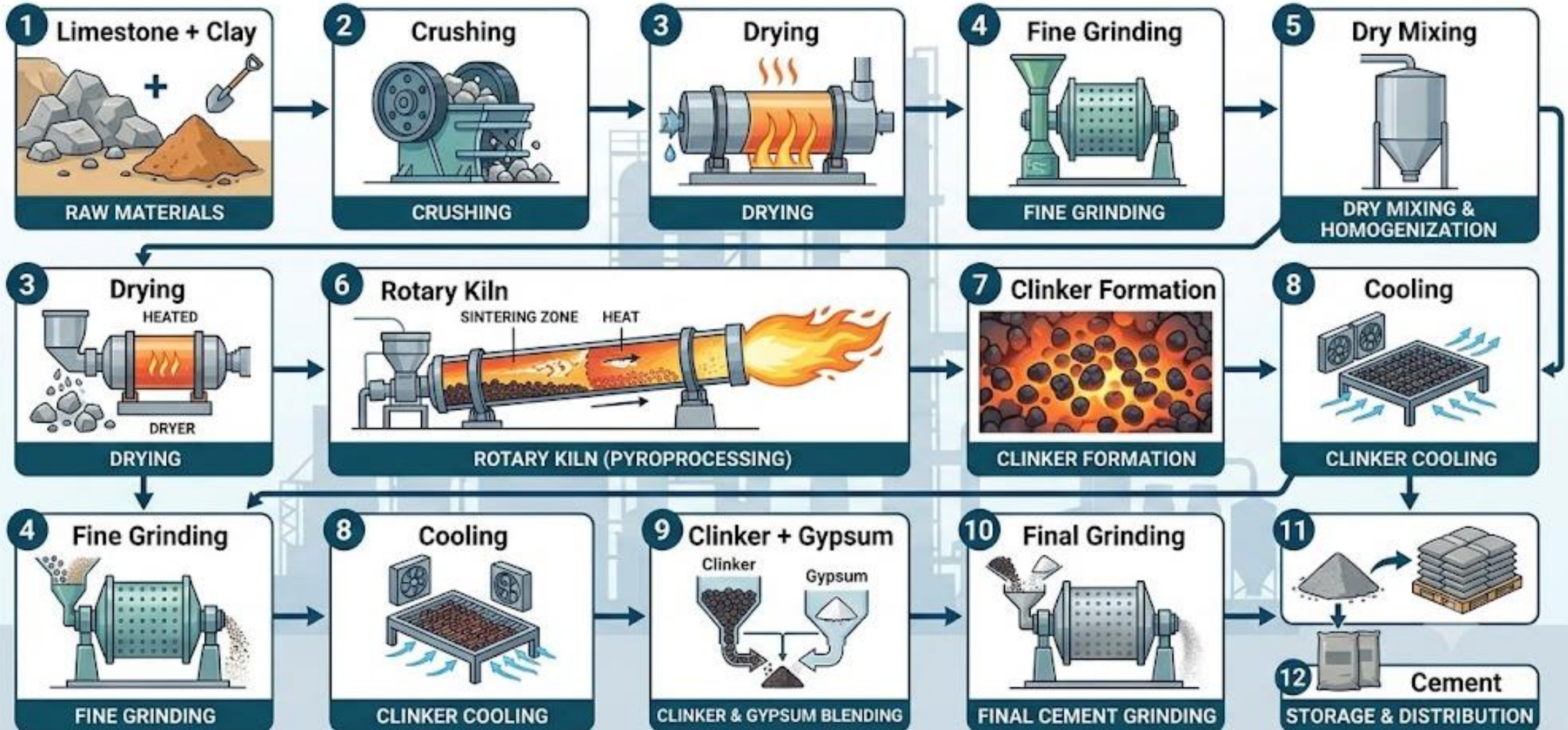
Hydration of Cement

- ❖ Cement gains its strength through a chemical reaction with water. This exothermic reaction between water and cement which liberates heat is known as hydration.
- ❖ The reaction by which cement acts as a bonding agent is called hydration of cement.
- ❖ This binding action takes place in the presence of water.



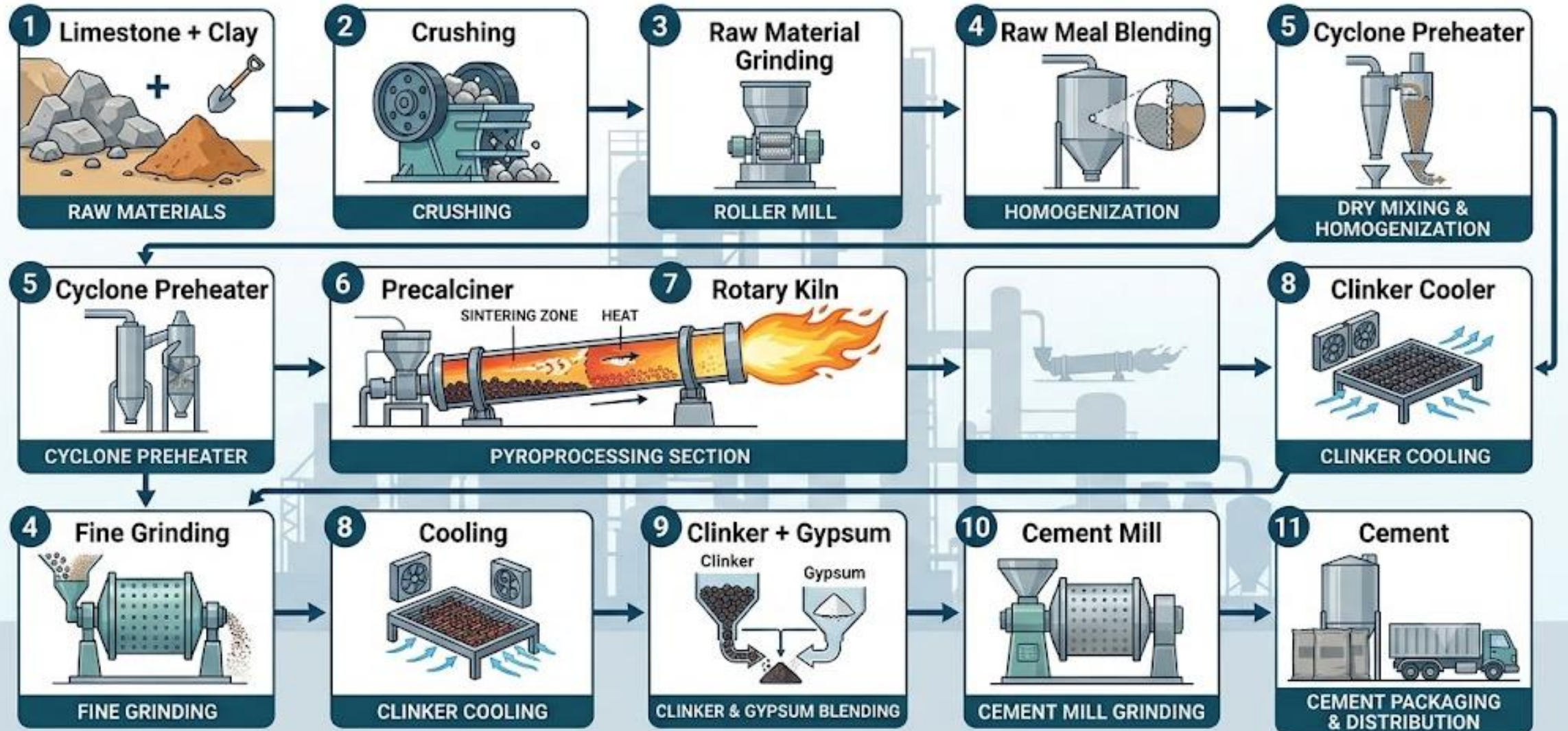


Manufacturing of Cement – Old Method





Manufacturing of Cement – New Method





Physical Properties of Cement

- ❖ Fineness – Gives idea about cement hydration & performance – **IS 4031 (Part 1)**
- ❖ Soundness – Determine volume stability of cement - **IS 4031 (Part 3)**
- ❖ Standard consistency – Amount of water required to make cement paste - **IS 4031 (Part 4)**
- ❖ Setting time – Find initial and final setting times of cement - **IS 4031 (Part 5)**
- ❖ Compressive strength – Find strength of cement mortar - **IS 4031 (Part 6)**
- ❖ Specific gravity – Find absolute volume of cement - **IS 4031 (Part 11)**



Field Tests on Cement

1. Colour test: Cement should possess uniform grey colour.

2. Physical property test:

1. Should feel smooth when rubbed in between fingers.
2. It should not feel oily when touched.
3. Cement should sink in water and should not float over the surface.
4. Sample should be free from presence of any lumps which are formed due to absorption of moisture.

3. Strength test:

1. Prepare a block of cement to be tested of size **25 × 25 × 200 cubic mm**.
2. Immerse in water for **7 days**.
3. Now remove the mould and subject it to point load of **340 N** by placing it over supports **150 mm apart**.
4. Sample should show no sign of failure under the application of this load.



Storing of Cement

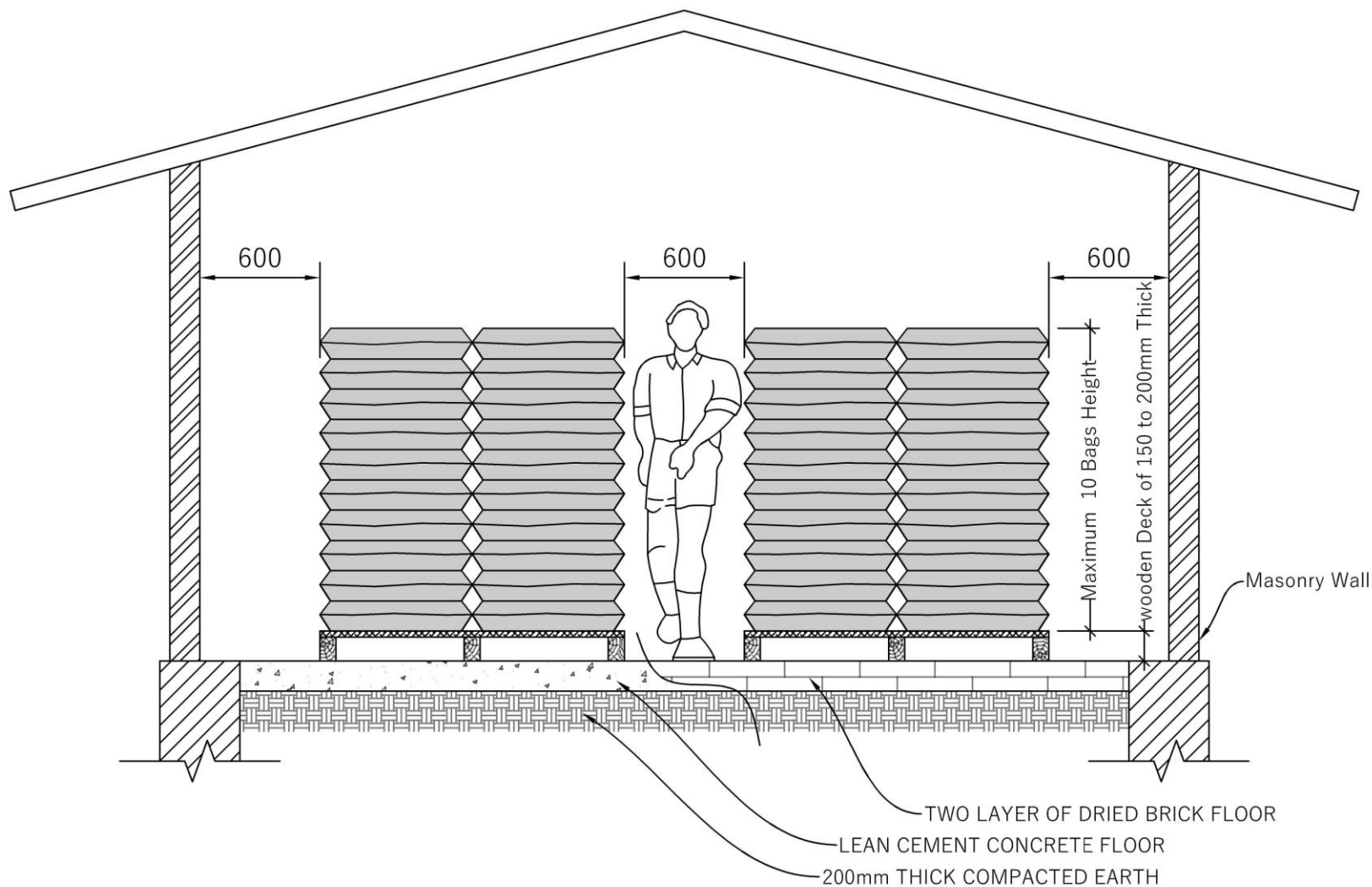
The cement should be stored in such a way that it cannot be exposed to the atmosphere. So, airtight bags are used to pack the cement. The absorption of moisture from the atmosphere will also depend upon the quality of cement.

Precautions:

- ✓ Cement should be stored in a special water-tight shed with a dry and damp-proof door, waterproof walls, and leak-proof roof.
- ✓ The stacks of cement bags should be placed at a distance of at least **30 cm from walls**.
- ✓ A stack should contain maximum **15 bags**.
- ✓ Width of a stack should not exceed **3 m**.
- ✓ Bags should be stacked on a raised platform, with sufficient air for movement between rows of stacks.
- ✓ Bags should be stored in such a way that **bags received first should be used first**.



Storing of Cement





Effect of Storage on Cement Properties

- Due to long storage, cement loses its strength, becomes lumpy and, due to its loss of capacity for hydration, becomes unusable.
- It is advisable to use cement within **3 months** at the maximum.
- The reduction in strength at different time periods is as follows:

Cement	Reduction in Strength
Fresh	NIL
After 3 months	15%
After 6 months	25%
After 1 year	40%
After 2 years	52%



Effect of Storage on Cement Properties

- ❖ Ordinary Portland Cement
- ❖ Low heat Cement
- ❖ Rapid Hardening Cement
- ❖ Portland Pozzolana Cement
- ❖ Sulphate Resisting Cement
- ❖ Blast Furnace Slag Cement
- ❖ White Cement
- ❖ Hydrophobic Cement
- ❖ Oil Well Cement

